Acceleration Questions

Calculating Acceleration

1. A lizard accelerates from 2m/s to 10m/s in 4 seconds. What is the lizard's acceleration?
2. A ball is dropped from the top of a building. After 2seconds, its velocity is measured to be 19.6 m/s. Calculate the acceleration of the dropped ball.
3. As a bus comes to a stop, it slows from 9m/s to 0m/s in 5 seconds. What is the acceleration of the bus?
4. During a race, a sprinter increases his speed from 5 m/s to 7.5 m/s over a period of 1.25 seconds. What is the sprinter's acceleration during this period?
5. A car advertisement states that a certain car can accelerate from rest to 70km/h in 7 seconds. Find the car's acceleration.
6. A cyclist accelerates from 0m/s to 8m/s in 3 seconds.
    a. What is his acceleration?
    b. Is his acceleration higher than that of a car which accelerates from 0m/s to 30m/s in 8 seconds?
7. A car manufacturer claims that its latest model can go from 0 to 90 km/h in 7.5 seconds.
    a. What is the car's final velocity in m/s?
    b. What is the car's acceleration in m/s^2?
8. A roller coaster car rapidly picks up speed as it rolls down a slope. As it starts down the slope, its speed is 4m/s. 3 seconds later, at the bottom of the slope, its speed is 22m/s. What is its acceleration?
9. You are travelling in a car that is moving at a velocity of 20m/s. You slam on your brakes and slow to 5m/s in 2 seconds. Calculate your acceleration.
10. A fighter jet landing on an aircraft carrier's flight deck that has a length of 300.0 meters must reduce its speed from about 42.5 m/s to exactly 0.0m/s in 2.0 seconds. What is the jet's acceleration?

Calculating Velocity

1. A cyclist accelerates at 2m/s^2 during a 5 second interval. What is the change in velocity of the cyclist?
2. If a Ferrari, with an initial velocity of 10 m/s, accelerates at a rate of 50 m/s^2 for 3 seconds, what will its final velocity be?
3. Josh rolled a bowling ball down a lane in 2.5 s. The ball travelled at a constant acceleration of 1.8 m/ s^2 down the lane and was travelling at a speed of 7.6 m/s by the time it reached the pins at the end of the lane. How fast was the ball going when it left Tim's hand?
4. The elevators in the Landmark Tower in Yokohama, Japan are among the fastest in the world. They accelerate upward at 3.125 m/s^2 for 4.0 seconds to reach their final velocity. If these elevators start at rest, what is their final velocity?
5. A freight train travelling with a velocity of 18.0 m/s to the south begins braking as it approaches the train yard. The train's acceleration while braking is -0.33m/s^2. What is the train's speed after 23 seconds?

Calculating Time

1. Falling objects drop with an average acceleration of 9.8 m/ s^2. If an object falls from a tall building, how long will it take before it reaches a speed of 49 m/s?
2.
3. A child sleds down a hill with an acceleration of 2.82 m/s^2. If her initial speed is 0.0 m/s and her final speed is 15.5 m/s, how long does it take her to travel from the top of the hill to the bottom?
4. The car with the fastest acceleration is the T*empest*. It has an acceleration of 6.89 m/s^2. Suppose the car accelerates from rest to a final speed of 26.8 m/s (about 96.5 km/hour). How long does it take the *Tempest* to reach this speed?
5. A remote control airplane is sitting on the runway and the operator revs the engine to full thrust then releases the brake. The plane accelerates down the runway at 1.6 m/s^2

and reaches its take off speed at 23m/s. How long did it take the plane to take off from the moment the brake was released?